PROJECT REPORT

UrbanCart — Customer Analytics & Customer Lifetime Value

Purchase analytics warehouse on the Olist Brazilian E-Commerce dataset
RFM · Historic CLV · Churn · Cohort Retention · Market Basket

Prepared by

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Stack: PostgreSQL star schema · Streamlit + Plotly dashboards · Power BI (PBIP) semantic model
Repository: github.com/kavyarathod05/urban-cart-customer-analytics-clv

Source: Olist Brazilian E-Commerce · CC BY-NC-SA 4.0 · analysis date 03 Sep 2018

1. Objective

Build an end-to-end customer analytics system for an e-commerce book: clean transactional data, a dimensional warehouse, RFM and historic CLV, inactivity-based churn, cohort retention, association rules, and five interactive dashboards — with Power BI as a mandatory second presentation layer. The original brief assumed omni-channel marketing data (campaigns, web logs, demographics, product cost). Those fields do not exist in a reliable public match, so the team selected the best real public dataset and redesigned scope around what the data can honestly support.

2. What We Did

2.1 Dataset decision

After comparing synthetic multi-table demos with real public sources, we standardised on the **Brazilian E-Commerce Public Dataset by Olist** (Kaggle, CC BY-NC-SA 4.0; orders Sep 2016 – Sep 2018). It supplies customers, orders, items, products, payments, and category translations — enough for RFM, CLV, churn, cohorts, and baskets. We explicitly **removed** campaign ROI, web-session funnels, gender/age, and cost/margin objects that Olist does not contain, rather than inventing them.

2.2 ETL and warehouse

Pipeline we implemented:

- **transform_olist.py** — join Olist CSVs into clean `customers / products / orders` extracts; map people to `customer_unique_id`; drop canceled/unavailable orders; amount = price + freight.
- **setup_local.py** — load PostgreSQL star schema (`dim_customers`, `dim_products`, `dim_date`, `orders_fact`) plus analytics views/tables for RFM, CLV, churn, cohorts, and association rules.
- **analysis_params.analysis_date** = `MAX(order_date)` (2018-09-03) so recency and churn are measured against the book's last day, not the machine's calendar.

2.3 Analytics models

RFM. R = analysis_date – last_order; F = COUNT DISTINCT order_id; M = Σ amount. R and M scored with NTILE(5); F scored 1/3/5 by rule.

```
CLV = AOV × purchase_frequency × lifespan_years
Bands (BRL): Low < 150 · Medium < 500 · High ≥ 500
```

Churn. High if inactive > 180 days, Medium > 90, else Low. Rate = High / buyers.

Cohorts. Cohort = month of first order; retention(c,p) = active in period p / cohort size. This board replaces marketing attribution (no ad spend in Olist).

```
Support(A,B) = pair_orders / total_orders
Confidence(A→B) = pair_orders / orders_with_A
Lift(A,B) = Support(A,B) / (Support(A) × Support(B))
```

2.4 Dashboards and Power BI

- **Streamlit + Plotly** (`app.py`) — five tabs with formula hover tips on every KPI and chart; custom theme (non-generic UI); Kaggle dataset link in the masthead.
- **Power BI** mandatory deliverable — `powerbi/export_dataset.py`, TMDL semantic model, `UrbanCart_Olist.pbip`, measures and page specification for Desktop (Windows).
- **Study guide PDF** — full-tab screenshots plus per-graph formulas for revision (`UrbanCart_Olist_Dashboard_Study_Guide.pdf`).

3. Deliverables Summary

Deliverable	What it contains
PostgreSQL warehouse	Star schema + RFM / CLV / churn / cohort / association objects
Streamlit app (5 tabs)	Executive, RFM, Churn, Cohorts, Product — live interactive views
Power BI PBIP	Import CSVs, semantic model, DAX measures, page layout spec
SQL scripts	RFM, CLV, star schema, DQ, incremental load, association, etc.
Documentation	README, Power BI README, study guide PDF, this project report

The following pages attach the exact dashboard screenshots from the live Streamlit app and describe what each board does and what the visuals show.

4. Dashboards — Screenshots & Work Done

Each figure is a full-tab capture of the Streamlit UI (analysis date 03 Sep 2018). Formulas appear as hover tips in the live app; key interpretations are summarised under each shot.

4.1 01 — Executive Overview (Ledger)

What we built. Built as the board-level health check for the Olist-backed book. KPIs: Gross GMV, Orders, Buyers, AOV, Avg CLV, and Churn Rate. Charts: monthly revenue area, orders-vs-buyers dual series, top-10 revenue by Brazilian state, and CLV band mix (Low / Medium / High).

What the board shows. About R$ 15.74M GMV across ~98k orders and ~95k buyers; AOV ≈ R$ 160. Most historic CLV mass sits in the Low band; São Paulo (SP) dominates state revenue. Orders and buyers track closely month-to-month — repeat purchase is thin in this marketplace extract.

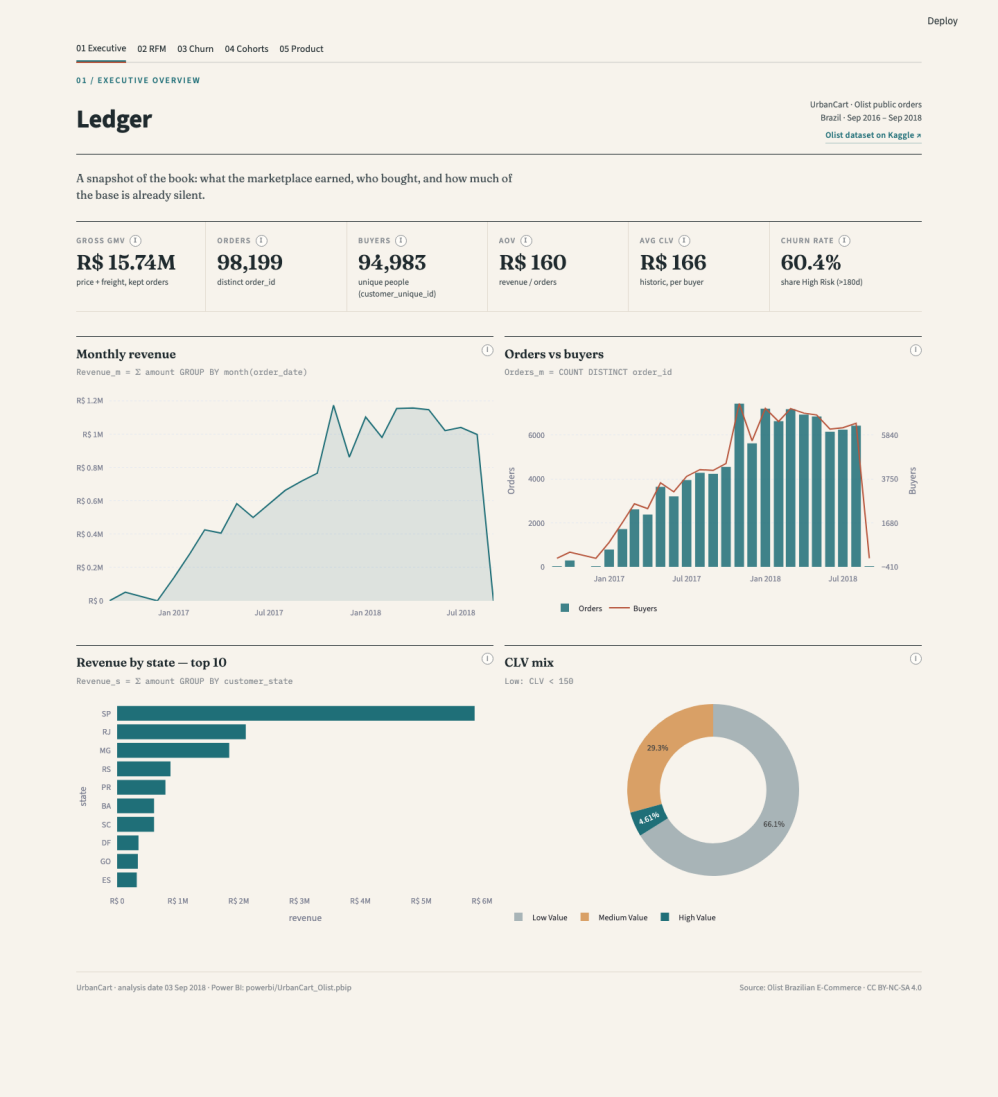


Figure 4.1 — 01 — Executive Overview (Ledger) (Streamlit · Olist)

4.2 02 — RFM Segmentation (Who still buys)

What we built. Implemented classic RFM with Olist-aware scoring: Recency and Monetary via NTILE(5); Frequency as rule scores 1 / 3 / 5 because most buyers order once (quintiles would collapse). Segments: Champions, Loyal, New, Average, At Risk, Lost. Visuals: headcount bars, revenue by segment, R×F heatmap, monetary box plots, and a summary table.

What the board shows. Headcount and GMV concentrate in New and Average customers. The R×F heatmap is heavy on F1 (single purchase) across all recency bins — the structural reason Frequency could not use NTILE.

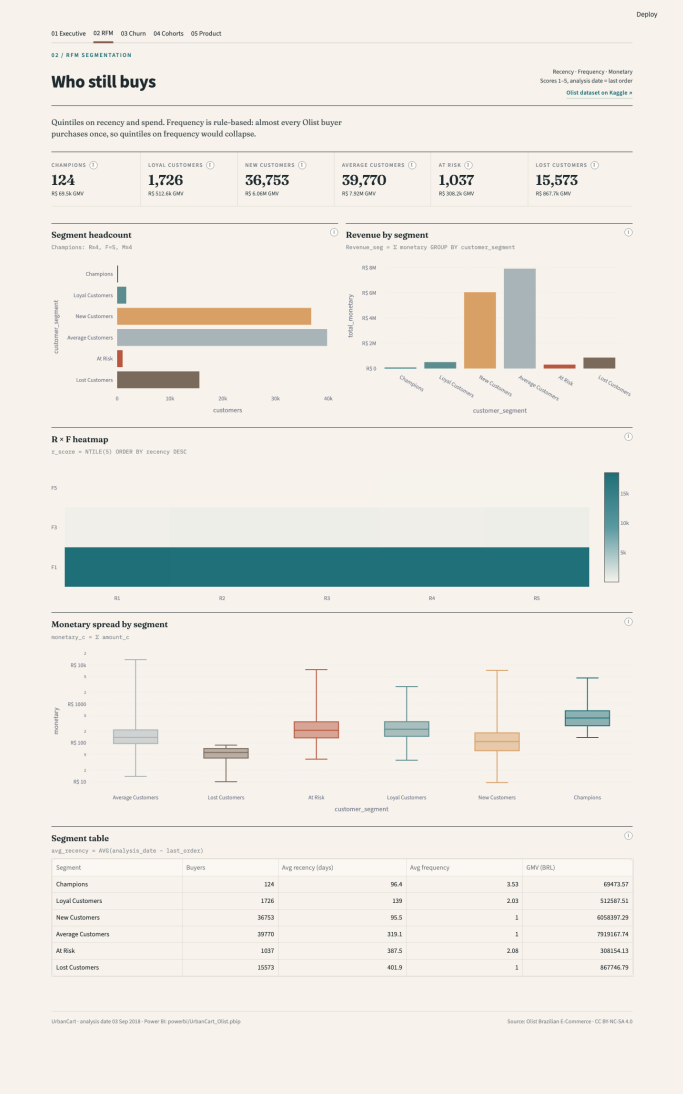


Figure 4.2 — 02 — RFM Segmentation (Who still buys) (Streamlit · Olist)

4.3 03 — Churn Risk (Who went quiet)

What we built. Defined inactivity churn against the dataset clock (analysis_date = MAX(order_date) = 2018-09-03), not today's calendar: High > 180 days, Medium > 90, else Low. Built risk mix donut, revenue sitting in each band, days-inactive histogram, risk by acquisition year, and a high-value high-risk buyer list for re-engagement study.

What the board shows. High Risk is ~60% of the base and carries the largest GMV share. Earlier acquisition years skew High Risk by construction of a fixed end date — an honest reading of a closed historical marketplace window.

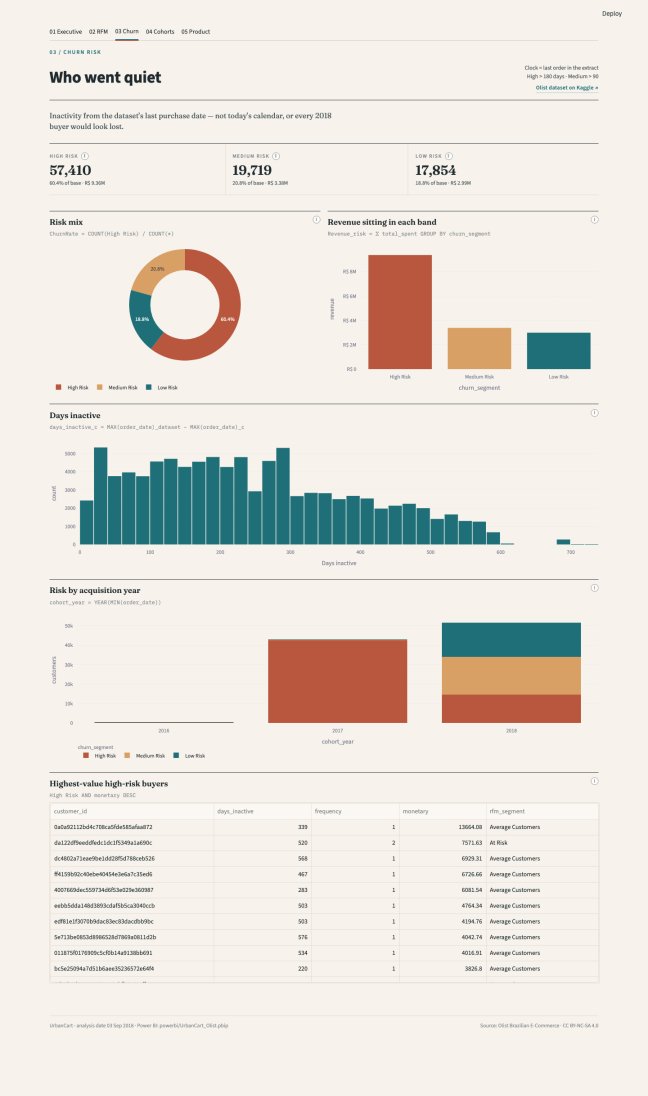


Figure 4.3 — 03 — Churn Risk (Who went quiet) (Streamlit · Olist)

4.4 04 — Cohorts & Retention (Who came back)

What we built. Replaced the original Marketing Attribution board (no ad-spend table in Olist) with first-order month cohorts. Built retention heatmap (period 0 = acquisition month), cohort-size bars, and revenue by months since first order. KPIs: repeat rate, cohort count, largest cohort.

What the board shows. Repeat rate ≈ 3%. Retention after M0 is very low; almost all revenue lands in period 0. Largest cohort around Nov 2017 (~7.2k buyers).

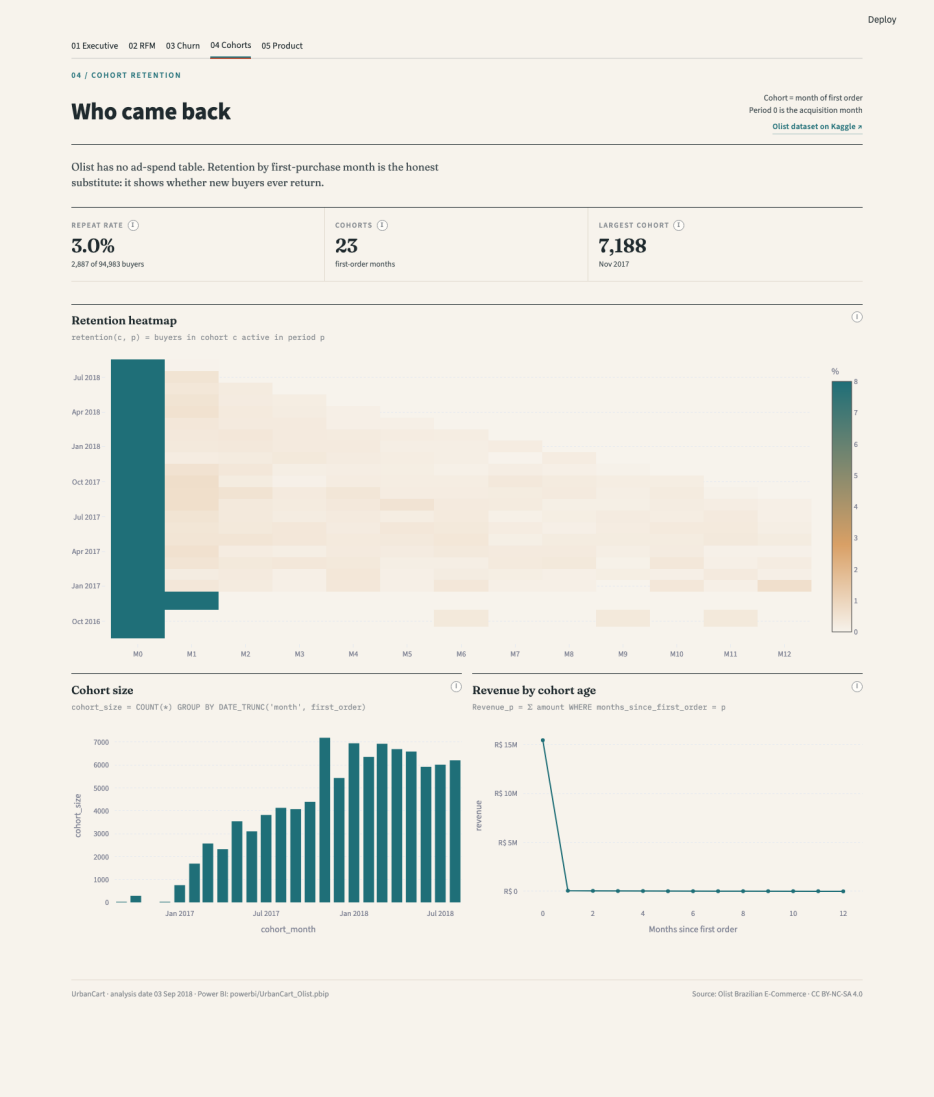


Figure 4.4 — 04 — Cohorts & Retention (Who came back) (Streamlit · Olist)

4.5 05 — Product & Affinity (What sold together)

What we built. Category treemap, top SKUs by revenue, units-vs-revenue bubble chart, category share over time, plus association rules at category and product grain (Support / Confidence / Lift on shared order_id baskets).

What the board shows. Health Beauty leads category revenue. Many baskets are single-category, so lift often sits near or below 1 — still useful for assortment hypotheses. Product-pair rules are sparse but high-signal where they appear.

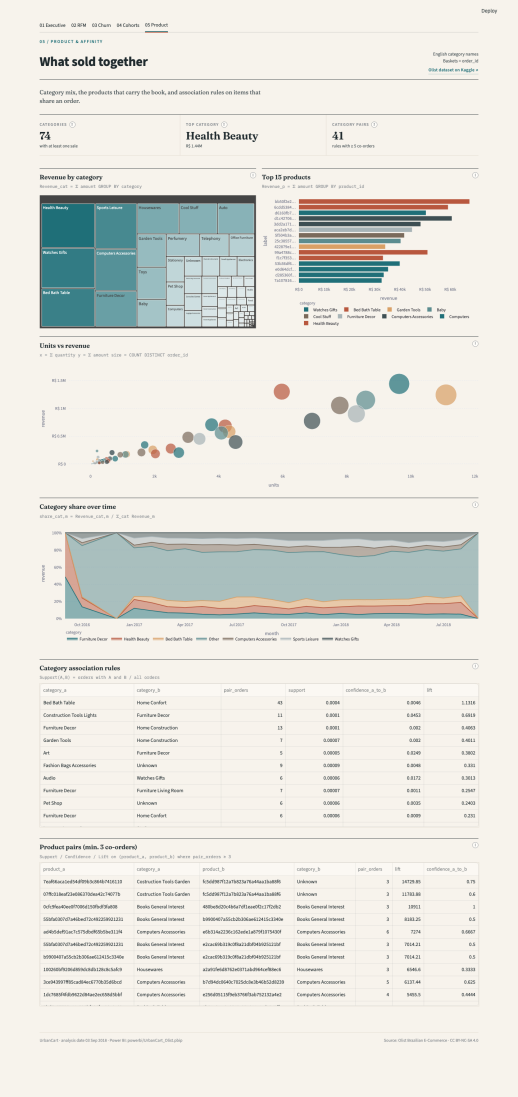


Figure 4.5 — 05 — Product & Affinity (What sold together) (Streamlit · Olist)

5. Key Outcomes

- Replaced synthetic / incomplete omni-channel assumptions with a real, citable public marketplace dataset and a scope that matches it.
- Delivered a reproducible ETL → Postgres → analytics → Streamlit + Power BI path, with analysis_date pinned to the data.
- Five coherent dashboards covering ledger health, RFM, churn, retention, and product affinity — each with transparent formulas.
- Documented findings that matter for study and defence: low repeat rate (~3%), High Risk majority by the fixed end date, SP-led geography, and sparse but interpretable association rules.

6. How to Re-run

Install PostgreSQL and Python 3.12, create database `urbancart`, place Olist CSVs under `dataset/raw/`, then:

```
python transform_olist.py
python setup_local.py
streamlit run app.py
python powerbi/export_dataset.py
python powerbi/generate_semantic_model.py
```

Open `powerbi/UrbanCart_Olist.pbip` in Power BI Desktop (Windows) and Refresh. Rebuild this report after re-capturing shots: `python docs/build_project_report.py`.

Team

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